Textbook

A Version-4 energy model of whole shopping baskets

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This is a joint probability model of a whole shopping basket, with an exact normaliser, fitted to transaction data and used for basket likelihood, basket generation, recommendation and price and promotion counterfactuals.

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1. Introduction

The model gives one probability to a whole basket chosen from thousands of products, computes that probability's normaliser exactly given a 16-dimensional latent vector, and uses the same law for likelihood, generation, recommendation and price and promotion counterfactuals.

1.1 What already exists

A shopping trip ends with a basket: a set of products bought together. Three families of models describe such baskets.

- **Multicategory choice models** from marketing (multivariate probit and multivariate logit) give a joint probability to the purchase or non-purchase of each of a few categories, with cross-category dependence and price effects. Their normaliser sums over every combination of categories, so they are used with a handful to a few dozen categories, not thousands of products.
- **Sequential models** such as SHOPPER treat a basket as products added one at a time, each choice conditioned on what is already in it. They scale to thousands of products and learn complements and substitutes. The probability of an unordered basket is a sum over all orderings, so it can only be estimated, by averaging over sampled orderings.
- **Representation and recommendation models** (product embeddings, basket completion, determinantal point processes) rank products or score sets. Most do not define a normalised probability for a whole basket that can be compared across models, and few include prices.

1.2 What this model adds

- **One normalised law over unordered baskets** from the full catalogue (5,455 products and 187 commodities on Dunnhumby), with product utilities that depend on household, store, week, price, discount, display, mailer and purchase history.

- **An exact normaliser given a latent vector.** Pairwise product interactions of rank 16 are removed by a Gaussian integral identity. Given the latent vector z, the sum over every allowed basket (up to 120 products) is computed exactly by elementary symmetric polynomials within each category and a dynamic program over basket size and number of categories. Exact purchase probabilities are its gradients.
- **Exact draws given z**, and a block Gibbs sampler on (basket, z) whose stationary law is the model.
- **Counterfactuals as the same law at a changed context** (another price, a display switched on), with a within-category competition term that lets a discount move purchases between rival products instead of only adding purchases.
- **Checks built in.** Every exact computation is tested against enumeration of all baskets of a small world. Synthetic worlds with a known causal truth test the counterfactuals, and an independent log-space calculation tests extreme cases.

1.3 How it can help

- **Comparable likelihoods.** Because the law is normalised over unordered baskets, held-out log-likelihood is a number other basket models can be scored against on the same trips.
- **Generation.** Synthetic baskets for simulation or for testing other methods, drawn from a fitted law rather than resampled from data.
- **Counterfactual questions** (what happens to this product, its rivals and the rest of the basket if its price falls) with the approximations stated and measured.
- **Open limits.** Section 7 lists what does not yet work: on Dunnhumby the discount response is too strong, and a promotion still adds a small gain to the rest of the basket that the data do not show.

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2. Related work

The model combines known pieces: an energy over subsets as in multivariate logit, the elementary-symmetric-polynomial and cardinality dynamic programs for exact subset sums, the Gaussian integral trick for pairwise terms, and stochastic maximum likelihood with persistent chains. What is new is putting them together for full baskets from a large catalogue with prices.

Family	Examples	Normalised probability of an unordered basket	Scale	Prices and promotions	Kinds of interaction
Multivariate probit and logit over categories	Manchanda, Ansari and Gupta 1999; Russell and Petersen 2000; Chib, Seetharaman and Strijnev 2002	yes, over category incidence	2 to 12 categories in these papers	yes	positive and negative, between categories
Sequential choice	SHOPPER, Ruiz, Athey and Blei 2020	only by summing over orderings	thousands of products	yes	complements and substitutes
Many categories, one product each	Donnelly, Ruiz, Blei and Athey 2021	per category	many categories	yes, with counterfactuals	through shared latent factors
Neural purchase prediction	Gabel and Timoshenko 2022	no, per-product probabilities	large assortments	yes (discounts, coupons)	learned by the network
Determinantal point processes	Gartrell, Paquet and Koenigstein 2017	yes	large catalogues	no	negative only (Kulesza and Taskar 2012)
Basket representations	Wan et al. 2018	no, rankings	large catalogues	no	complementarity in embeddings
This model	—	**yes, exact given z**	**5,455 products, 187 commodities**	**yes**	**positive and negative pairs, rank 16**

2.1 Multicategory choice models

Multivariate probit and logit models describe whether a household buys in each of several categories on a trip, with correlated preferences or cross-category terms. Manchanda, Ansari and Gupta (1999) and Chib, Seetharaman and Strijnev (2002, twelve categories) use a multivariate probit. Russell and Petersen (2000) build the joint law of a basket from conditional logit models for each item, which is the form our pseudo-likelihood stage also uses. These models give a normalised joint law, but the

normaliser sums over every combination of categories, so the number of categories stays small, and they describe category incidence, not which products are bought.

2.2 Sequential and factorisation models

SHOPPER (Ruiz, Athey and Blei 2020) models a trip as a sequence of choices, each depending on the items already chosen, with latent product and household factors and price sensitivities. It handles thousands of products and answers price-change questions. Because its probabilities are for ordered sequences, the probability of an unordered basket has to be estimated by averaging over orderings; in this project it is scored with 8,192 sampled orderings. Donnelly, Ruiz, Blei and Athey (2021) model choices in many categories at once with matrix factorisation, with at most one product per category, and use the model for counterfactual discounts. Gabel and Timoshenko (2022) predict each product's purchase probability with a neural network that scales to large assortments and evaluate it on experimental price discounts.

2.3 Set and basket models from machine learning

Determinantal point processes give an exactly normalised law over subsets and have been used for basket completion (Gartrell, Paquet and Koenigstein 2017). As joint laws over item inclusion they can express only negative correlation between items (Kulesza and Taskar 2012), so they cannot represent complements bought together. Representation models such as Wan et al. (2018) learn product and user embeddings from co-purchase and use them to recommend, without a normalised basket probability.

2.4 Methods this model builds on

- **Exact sums over subsets of a fixed size.** The total weight of all k-subsets of a category is the elementary symmetric polynomial of the weights, the same quantity behind the conditional Bernoulli (maximum-entropy fixed-size sampling) design of Chen, Dempster and Liu 1994 (cited there). Exact inference with potentials on how many variables are on is studied by Tarlow, Swersky, Zemel, Adams and Frey 2012; our size dynamic program is of that kind, extended to count products and categories together.
- **Gaussian integral trick.** Writing exp of a quadratic as an expectation over a Gaussian vector turns pairwise binary interactions into independent terms given the vector (Zhang, Ghahramani, Storkey and Sutton 2012). Here it removes the rank-16 pair term so the exact program applies given z.
- **Estimation of energy models.** Maximum pseudo-likelihood (Besag 1975) gives a fast first fit from exact one-product conditionals. Stochastic maximum likelihood with persistent Markov chains (Younes 1989; Tieleman 2008) estimates the gradient of log Z from the model's own samples.
- **Within-category competition.** The competition term uses the category's inclusive value (log-sum of utilities) and a dissimilarity parameter between 0 and 1 in the way of a nested logit (Train 2009). The parameter is estimated with an instrument built from price variation, in the spirit of instrumental-variable demand estimation (Berry 1994).

2.5 Data

The real data are dunnhumby's "The Complete Journey": two years of in-store purchases of about 2,500 households, with product, campaign and in-store promotion (display and mailer) tables (dunnhumby).

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3. Methodology I: the problem

The task is to fit one probability law over the baskets a household can buy on a trip, given everything known about the trip, such that its normaliser and its draws can be computed and it can be re-evaluated at prices and promotions that were not observed.

3.1 Data and notation

- A **trip** t is one visit of household h to store s on day d (week w). The data hold 158,983 training trips on Dunnhumby (weeks up to 82); weeks 83–90 are for validation and weeks 91–101 for the test, opened once per final model.
- The store offers products $\mathcal{J}_t$; on Dunnhumby the whole training catalogue, J = 5,455 products.
- Products are partitioned into **categories** c = 1, ..., C (C = 187 commodities); $\mathcal{J}_{tc}$ are the offered products of category c.
- The **basket** is a set $S \subseteq \mathcal{J}_t$. Its size is n, its count in category c is n_c, and its number of categories present is m.
- The **context** x_t holds the household, store, week, each product's price and regular price, display and mailer flags, and the household's purchase history strictly before day d.

$$n = |S|, \qquad n_c = |S \cap \mathcal{J}_{tc}|, \qquad m = \#\{c : n_c > 0\}$$

3.2 The support

The model puts probability only on non-empty baskets no larger than the largest training basket (n_max = 120) and with at most R products from one category (R = 40 in the current model):

$$\mathcal{S}_t = \{\, S \subseteq \mathcal{J}_t : 1 \le n \le n_{\max},\ n_c \le R \text{ for all } c \,\}$$

The size cap is part of the law, not only a computational device: the pair term below grows with the square of the basket size, and without the cap it can put most of the probability on baskets far larger than any observed (on Dunnhumby, a Gibbs chain without it went from 36 to about 2,660 products in 100 sweeps).

3.3 The law to be fitted

The model is an energy-based law: every basket gets an energy E, and its probability is proportional to exp E.

$$p_\theta(S \mid x_t) = \frac{\exp E_\theta(S, x_t)}{Z_\theta(x_t)}, \qquad Z_\theta(x_t) = \sum_{A \in \mathcal{S}_t} \exp E_\theta(A, x_t)$$

The energy (section 5) is a sum of product utilities, pairwise interaction terms, within-category count terms and a size term. The parameters θ are fitted by maximising the log-likelihood of the observed training baskets:

$$\hat{\theta} = \arg\max_\theta \sum_{t \in \text{train}} \Big[E_\theta(S_t, x_t) - \log Z_\theta(x_t) \Big] \; - \; \text{penalties}(\theta)$$

Settings and model variants are chosen on validation log-likelihood; the test weeks are scored once.

3.4 What the solution must deliver

1. **Likelihood.** log p(S | x) for any basket and trip, so models can be compared on held-out baskets.
2. **Generation.** Draws of whole baskets from p(· | x).
3. **Conditionals.** The probability of each product given the rest of the basket (recommendation).
4. **Counterfactuals.** p(· | x′) for a changed context x′ (a price, a discount, a display), from the same parameters.

3.5 Why it is hard

The normaliser is a sum over every allowed basket. With 5,455 offered products and baskets of up to 120, the number of terms is of the order of C(5455, 120), far beyond enumeration. Pairwise interactions between products couple every product with every other, so the sum does not factor. Section 4 shows how the sum is made exact given a low-dimensional latent vector.

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4. Methodology II: how it is solved

The pair term is removed by a Gaussian latent vector z; given z the normaliser is an exact polynomial-time sum, exact draws follow from the same tables, and the remaining integral over z is handled by Gibbs sampling (for draws and training) and importance sampling (for likelihood).

4.1 Removing the pair term with a Gaussian latent

Each product carries a vector ϕ_j of dimension r = 16. The pair term is ½‖s(S)‖² minus a per-product part, where s(S) is the basket's summed vector (section 5.3 gives its exact form). For any vector s the Gaussian identity holds:

$$e^{\frac{1}{2}\|s\|^2} = \mathbb{E}_{z\sim N(0,I_r)}\, e^{z^\top s}$$

Applied to the energy, it gives a law in which, for fixed z, products interact only through their category counts and the size term:

$$e^{E(S)} = \mathbb{E}_{z\sim N(0,I)}\, e^{E_z(S)}, \qquad u_{tj}(z) = b_{tj} - \tfrac{1}{2}\|\phi_j\|^2 + z^\top \phi_j$$

E_z(S) is the energy with b replaced by u(z) and no pair term. Then Z = E_z[Z_z], and the joint law of basket and latent has simple conditionals:

$$p(S,z) \propto N(z;0,I)\, e^{E_z(S)}, \qquad z \mid S \sim N\big(s(S), I\big), \qquad p(S \mid z) = e^{E_z(S)}/Z_z$$

The basket marginal of p(S, z) is exactly the model p(S).

4.2 The exact normaliser given z

Within a category. With weights $w_j = \exp u_{tj}(z)$, the total weight of all k-product subsets of category c is the elementary symmetric polynomial e_{c,k}, computed by a recursion over the category's products with no subtraction:

$$e_k^{(i)} = e_k^{(i-1)} + w_i\, e_{k-1}^{(i-1)}, \qquad k \le R$$

Every other term of E_z depends on the basket only through n_c, so category c contributes a polynomial with coefficients

$$g_{c,k} = e_{c,k}\, \exp\Big(-\rho_c\tbinom{k}{2} - f_c(k) - (k-1)_+\, a_{tc}(z)\Big), \qquad g_{c,0} = 1$$

Across categories. The size term acts on the number of categories present (section 5.5), so the product over categories tracks both the number of products n and the number of categories m:

$$H_c(y,v) = 1 + v\sum_{k\ge 1} g_{c,k}\, y^k, \qquad Z_z = \sum_{n=1}^{n_{\max}} \sum_{m=1}^{n} e^{-\rho_0(m)}\, [y^n v^m] \prod_c H_c(y,v)$$

The product is built category by category in an (n, m) table, truncated at n_max. The ESPs cost O(|$\mathcal{J}_t$| R) per trip and the table O(C_t n_max² R).

Purchase probabilities are gradients. Because $u_{t\,j}$ enters E_z linearly with coefficient 1[j ∈ S], the gradient of log Z_z with respect to $u_{t\,j}$ is the exact probability that product j is in the basket given z. The backward passes are written by hand in C++.

Numerics. Weights of different basket sizes can differ by far more than a double can hold (one product with utility 19 next to ordinary ones makes size 120 about e^1000 heavier than size 1). Every row n of the table therefore keeps its own power-of-two exponent. An earlier version with a single scale per trip rounded large sizes to zero and lost up to 123 nats of log Z on 5 of 256 real trips at chain-end z; it was fixed on 2026-10-26 and is tested against enumeration.

Per-trip item cap. Given z, most trips put almost no weight on large baskets. For each (trip, z) the sampler chooses a cap K ≤ n_max with a proven bound that the baskets above K carry at most 10^{-8} of the mass, and runs the table only to K. The bound uses a tilted sum over all categories without the size cap; K grows to n_max whenever z needs it.

4.3 Exact draws given z

The same tables give exact draws by backward sampling, in three steps:

1. draw (n, m) with probability proportional to exp(−ρ_o(m)) times its table entry;
2. for each category from last to first, draw its count n_c given the products and categories still to place;
3. within the category, go through its products from last to first and include product i with the probability below, where k′ is the number still needed.

$$P(i \text{ included}) = \frac{w_i \, e^{(i-1)}_{k'-1}}{e^{(i)}_{k'}}$$

Each step is an exact conditional, so the draw is exact.

4.4 Draws from the full model: block Gibbs

Alternating the two conditionals of section 4.1 is a Gibbs sampler on (S, z) whose stationary law is p(S, z), so its baskets are draws from the model:

$$z^{(i)} \sim N\big(s(S^{(i-1)}), I\big), \qquad S^{(i)} \sim p\big(\cdot \mid z^{(i)}\big)$$

Given any z every allowed basket has positive probability, so the chain is irreducible on the support. Chains start from an exact draw of the model without interactions and can be tempered (pair term

multiplied by β ∈ [0, 1]).

4.5 The likelihood: importance sampling over z

log p(S) = E(S) − log Z needs log Z = log E_{N(0,I)} Z_z, a 16-dimensional integral. It is estimated by importance sampling with a proposal built from the model's own baskets:

$$\widehat{Z} = \frac{1}{D}\sum_{d=1}^{D}\frac{N(z_d;0,I)\,Z_{z_d}}{\pi(z_d)}, \qquad \pi(z) = (1-\epsilon)\frac{1}{K}\sum_{k=1}^{K} N(z;s_k,I) + \epsilon\,N(z;0,\Sigma)$$

- The centres s_k are basket vectors s(S) from four chains: one from β = 0, and three started apart and annealed through β = 0.25, 0.5, 0.75, 1. The z-marginal of the model is exactly a mixture of N(s(S), I) over its baskets, so this proposal approximates it.
- A wide defensive component (ε = 0.1) bounds the weights in the tails.
- Each half of the draws uses its own stratified component picks; the difference of the two half-estimates and the effective sample size are reported with every estimate.
- log Ẑ is consistent but biased low for finite D (Jensen's inequality).

4.6 Estimation

Training runs in stages, each starting from the previous one.

1. **Base model** (no pair term, size term on n): exact maximum likelihood with AdamW, plus a size-law penalty, an aggregate price-elasticity anchor and a pooling penalty on factor products.
2. **Pseudo-likelihood** (one epoch). The probability of each offered product given the rest of the basket is an exact logistic function of the energy difference $d_j = E(S \cup j) - E(S \setminus j)$; the pseudo-likelihood sums its log over every offered product of every trip (Besag 1975).
3. **Stochastic maximum likelihood** (up to 6 epochs). The gradient of log Z is an expectation over the model's z-marginal of the exact gradient given z:

$$\nabla_\theta \log p(S) = \nabla_\theta E(S) - \mathbb{E}_{q(z)}\left[\nabla_\theta \log Z_z\right]$$

Each training trip keeps a persistent Gibbs chain started at its observed basket. Each step runs 4 sweeps, draws z ~ N(s(S_chain), I) and steps on $E(S_t) - \log Z_z$ with the exact program and its gradient. The only approximation is the chain's distance from stationarity. Parameters are averaged (Polyak) after the first third of the stage, and validation log-likelihood on 1,024 held-out training trips selects among them.

4. **Calibration** (one epoch): the same steps with the pair vectors and context factors fixed.

Throughout, size and category terms also take moment-matching steps (model frequency minus data frequency, scaled by the data frequency), priors are added, and gauge directions the likelihood cannot see are fixed by centring. The competition parameters λ_c are not learned by gradient: an instrumental-variable estimator (section 5.2) re-estimates them before training, after every epoch and at the end.

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5. Model components

The energy of a basket is the sum of its products' utilities, a pairwise interaction term, within-category count terms and a size term on the number of categories present; every term except the pair term depends on the basket only through which products and how many per category, which is what keeps the normaliser exact given z.

$$E(S) = \sum_{j \in S} \left(b_{tj} + \iota_{t\,c(j)} \right) \;+\; \mathcal{P}(S) \;-\; \sum_{c} \left[\rho_c \binom{n_c}{2} + f_c(n_c) \right] \;-\; \rho_0(m)$$

Component	What it captures	Parameters (Dunnhumby commodity model)
Popularity λ_j	baseline attraction of each product	5,455
Taste $\theta_h^\top \alpha_j$	household × product preference, rank 32 incl. a household size shift	1,920 × 32 and 5,455 × 32
Taste per category $\theta_h^\top u_c$ (M4a)	household × category preference	187 × 32
Price slope g_{hj}	household × product price sensitivity, rank 8, non-negative	1,920 × 8 and 5,455 × 8
Relative price κ, discount split κ_D, c_D	response to relative price and to discount depth	3
Display and mailer	one common slope each plus per-product deviations shrunk to zero	2 + 2 × 5,455
Season $\mu_j^\top \delta_w$, store $\zeta_j^\top \xi_s$	week-of-year and store effects per product	8 factors over 53 weeks; 4 factors over 115 stores
Recency $\psi_j^\top r_{tj}$ and product history ψ_c^⊤r^item	timing of the household's past purchases	5,455 × 4 and 187 × 3
History centring and size term	keeps history from inflating basket size	1
Competition λ_c	within-category switching under price and promotion changes	187
Pair vectors $\phi_j = \psi_c + \delta_j$ (M2)	complements and substitutes across products and categories, rank 16	5,455 × 16 (incl. 187 × 16 category vectors)
Count terms ρ_c, f_c (M1)	several products from one category	187 and 187 × 3
Size term $\rho_0(m)$ (M3)	number of categories in the basket	120

5.1 Product utility

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\begin{aligned}
b_{tj} = {} & \lambda_j + \theta_h^\top \alpha_j + \theta_h^\top u_{c(j)} && \text{popularity, taste} \\
& - g_{hj}\left[\bar{m}_t + \kappa(\Delta_{tj} - \bar{m}_t)\right] - g_{hj}(\kappa_D - \kappa) D_{tj} - c_D D_{tj} && \text{price, discount split} \\
& + (w^{\mathrm{dsp}} + \omega_j^{\mathrm{dsp}})\,\mathrm{dsp}_{tj} + (w^{\mathrm{mlr}} + \omega_j^{\mathrm{mlr}})\,\mathrm{mlr}_{tj} && \text{display, mailer} \\
& + \mu_j^\top \delta_w + \zeta_j^\top \xi_s && \text{season, store} \\
& + \psi_j^\top r_{tj} + \psi_{c(j)}^{\mathrm{item}\top} r_{tj}^{\mathrm{item}} - o_t && \text{purchase history, centred}
\end{aligned}
$$

- **Taste.** θ_h and α_j have 32 coordinates; the last coordinate of α is fixed at 1, so the last coordinate of θ_h shifts every product of household h equally (a household basket-size shift). The other loadings are centred over products.
- **Price.** Δ_{tj} is the log chain-wide weekly price minus the product's training-mean log price; $\bar{m}_t$ is its average over the trip's assortment. The slope $g_{hj} = \Sigma_q \text{softplus}(\gamma_h q)\,\text{softplus}(\beta_j q)$ is non-negative, so a higher price never raises utility. κ scales the response to a price relative to the rest of the assortment.
- **Discount split.** $D_{tj} = \min(\log(\text{price} / \text{regular price}), 0)$ is the discount depth. It gets its own multiplier κ_D and a slope c_D common to all households and products. With $\kappa_D = \kappa$ and $c_D = 0$ this is the one-slope model, where training starts.
- **Display and mailer.** 0/1 flags per product, store and week. Each has one common slope plus a per-product deviation with a prior that shrinks it to zero, so rarely displayed products do not get extreme weights.
- **Purchase history.** r_{tj} (4 features: never bought in the sub-commodity, and three decays of the days since it last did) and r^item (3 features: bought this product before, it was the latest choice in its category, decay since last bought). All are computed strictly before the trip.
- **History centring.** History terms only grow over time, and raising many utilities raises basket size. So every utility of a trip is shifted by the same o_t = (mean history term over the trip's assortment) + $\eta \log(1 + \text{products ever bought})$. History then decides which products, and η is the only way it changes how many.

Every term is a fixed function of the context, so utilities never depend on the basket.

5.2 Within-category competition

A discount raises a product's utility. In a law of whole baskets that also raises the chance that its category is bought at all, and real discounts mostly move purchases between rival products. On Dunnhumby about 90% of the effect is switching within the category. The competition term shifts every product of category c on trip t by

$$\iota_{tc} = (\lambda_c - 1)\big[\mathrm{IV}_{tc}(b) - \mathrm{IV}_{tc}(\bar{b})\big], \qquad \mathrm{IV}_{tc}(b) = \log \sum_{j \in \mathcal{J}_{tc}} e^{b_{tj}}, \qquad \lambda_c \in (0, 1)$$

where $\bar{b}$ is the trip's utility at regular prices (no price deviation, discount, display or mailer). The category's inclusive value then rises by only λ_c times its gain, as in a nested logit with dissimilarity λ_c: λ_c near 0 is pure switching, $\lambda_c = 1$ removes the term. ι depends on the context only, so the normaliser stays exact.

Likelihood barely identifies λ_c: across an 18-fold range it changes the log-likelihood by less than 0.002 nats per trip, and purchase history correlates with promoted items. So λ_c is estimated by an instrument.

1. **The instrument.** A price shock per (trip, category) built only from store-week prices, discounts and population-average slopes, weighted by popularity shares. It is centred within category and within trip. Display and mailer are left out, because their per-product weights are fitted on the same trips.

2. **The first stage.** The model's exact expected category count at λ_c = 1 and at λ_c = 0, from the exact program.

3. **The ratio.** Projecting both the observed counts and this exact first stage on the instrument gives a Wald ratio per category, shrunk towards the pooled ratio and clipped to [0.01, 0.99], with a category-jackknife standard error.

On the commodity model the pooled raw estimate at the start is 1.51 ± 0.31 (clipped to 0.99); after training λ_c averages 0.34 (range 0.17–0.48).

5.3 Pairwise interaction (M2)

Each product has a vector $\phi_j = \psi_{c(j)} + \delta_j$ of dimension 16: a category part and a product part. The category part counts once per category present, so a pair of categories that go together raises the probability of buying from both without also rewarding many products from each:

$$s(S) = \sum_{c:\, n_c > 0} \psi_c + \sum_{j \in S} \delta_j, \qquad \mathcal{P}(S) = \tfrac{1}{2}\|s(S)\|^2 - \tfrac{1}{2}\sum_{j \in S}\|\phi_j\|^2 + \tfrac{1}{2}\sum_{c}(n_c - 1)_+ \|\psi_c\|^2$$

Positive inner products $\phi_i^{\mathsf{T}}\phi_j$ are complements, negative ones substitutes. The rank-16 form is what lets the Gaussian identity of section 4.1 remove the term with a 16-dimensional z.

5.4 Count terms within a category (M1)

ρ_c C(n_c, 2) penalises (ρ_c > 0) or rewards (ρ_c < 0) several products from one category, and f_c gives free adjustments at 2, 3 and 4 or more products. Safety bounds keep any category from rewarding a basket by more than 1.5 nats, and the cap R = 40 limits products per category; on the earlier Dunnhumby model at most 4.5×10^{-10} of a trip's probability would lie above that cap.

5.5 Size term (M3)

The size term $-\rho_0(m)$ acts on the number of categories present, not on the number of products; how many products per category is governed by ρ_c and f_c. $\rho_0(0) = 0$, ρ_0 is a free function of m = 1, …, 120, and baskets are non-empty. Section 7 reports that in the fit trained before the 2026-10-26 normaliser fix, ρ_0 rises to m = 9 and then falls, which rewards baskets with many categories.

6. Applications

Every application below is a computation on the one fitted law: a counterfactual is the law at a changed context, a generated basket is a draw from it, a recommendation is one of its conditionals, and a likelihood is its value at an observed basket.

Application	What is computed	Exact or approximate
Price and promotion counterfactuals	purchase probabilities of every product at a changed context	exact given z; over z by reweighting chain draws
Basket generation	whole baskets drawn from the law	exact given z; Gibbs chain over z
Recommendation	probability of adding each product given the rest of the basket	exact
Likelihood and model comparison	log p(S) of held-out baskets	exact given z; over z by importance sampling
Learned structure	complements, substitutes, tastes, price sensitivities, segments	read from parameters
Promotion environment	weekly revenue under a discount policy	uses the counterfactual machinery

6.1 Price and promotion counterfactuals

A counterfactual changes the context x to x′: a product's price, a discount depth, a display or mailer flag. Changing product k's log price by Δ changes its $\Delta_t k$, moves the trip's mean price $\bar{m}_t$ by $\Delta / |\mathcal{J}_t|$, changes the discount depth when there is a discount panel, and changes the competition shift ι of its category.

Base model (no pair term). Recompute utilities at x′ and take the gradient of log Z: the new purchase probabilities are exact.

Version 4. The z-marginal changes with the context. Latent draws z_k taken from chains at the factual context are reweighted by Z′_z / Z_z (self-normalised importance weights), and each term is the exact probability given z:

$$\widehat{P}'(j \in S) = \sum_k \omega_k \, P'_{z_k}(j \in S), \qquad \omega_k = \frac{Z'_{z_k} / Z_{z_k}}{\sum_l Z'_{z_l} / Z_{z_l}}$$

One set of chains serves every scenario, so differences between scenarios carry no independent Monte Carlo noise. An option holds each trip's latent state fixed (equal weights), so only the basket given z responds.

One utility changes. If only product j's utility changes by Δ, the result is exact for every z and the reweighted marginal is the same tilt of the factual marginal $\bar{p}$:

$$P'(j \in S) = \frac{\bar{p}\, e^{\Delta}}{1 - \bar{p} + \bar{p}\, e^{\Delta}}$$

What is reported. The change in the product's own purchases, in its rivals' (same category), and in the rest of the basket. A real discount should raise the first, lower the second, and leave the third near zero. The discount check compares the model's discount-depth slope and display and mailer effects with the same quantities measured directly on discount events in the data.

6.2 Basket generation

To generate the basket of a trip:

1. build its context (prices, promotions, household, history) and utilities;
2. draw a starting basket exactly from the model without interactions (β = 0);
3. run Gibbs sweeps at β = 1 (200 in evaluation);
4. return the final basket.

A counterfactual basket is generated the same way from the changed context. Expected purchases are estimated without sampling noise from the basket draws by averaging exact purchase probabilities over the chain's z values (Rao–Blackwellisation).

Realism check. One basket is generated per test trip and seed. A gradient-boosted classifier is trained to tell real from generated baskets, with folds grouped by household. Its held-out AUC measures realism: 0.5 means indistinguishable. A second check compares the model's expected basket size with the real size in each period (early and late training weeks, validation, test).

6.3 Recommendation

The probability of adding product j to a basket given the rest is exact and cheap, because only the energy difference matters:

$$\log \frac{P(j \in S \mid S \setminus j)}{P(j \notin S \mid S \setminus j)} = E(S \cup j) - E(S \setminus j)$$

It includes the product's utility for this household and trip (price, promotion, history), its pair terms with the products already in the basket, and the change in the category and size terms. The check hides one product of a validation basket and ranks every offered product by this log-odds. On the earlier co-purchase-partition model (1,620 cases) the mean reciprocal rank was 0.118 and recall at 10 was 21.8%, against 0.093 and 18.0% for the previous model and 0.045 for popularity.

6.4 Likelihood and model comparison

Because the law is normalised over unordered baskets, held-out log-likelihood per basket compares models directly on the same trips. Comparisons are paired by trip, with household-clustered intervals on the test weeks. The estimate reports its effective sample size and half-split difference, so a poorly converged estimate is visible. SHOPPER is scored on the same trips by averaging over 8,192 orderings.

6.5 What the model learns

On the earlier Dunnhumby Version-4 model:

- **Interactions.** The first pair direction carries 59% of the interaction variance, and the category part 87%. Learned pair strength correlates 0.40 with real log co-purchase lift.
- **Taste.** 36% of each product's 5 nearest neighbours in taste space share its commodity (1.6% by chance). The household size shift is higher for large households and households with children.
- **Price.** A household's price slope correlates 0.60 with the share it actually buys on discount.
- **Segments.** k-means on taste, size shift and price slope gives five segments whose over-bought commodities in training weeks correlate 0.67–0.94 with those in validation weeks. There are no sharp clusters (silhouette about 0.02).

6.6 Promotion environment

A Gymnasium environment: one step is one week, the action is a discount of 0, 10, 20 or 30% for each of the 12 highest-revenue commodities, and the reward is expected revenue at paid prices relative to the same week without promotion. It replays a fixed sample of that week's real trips (500 by default) and uses either the base model or Version 4. On synthetic worlds it can be scored against the true revenue. Limits: a promotion cannot bring extra shoppers to the store, units per product are training averages, and simulated purchases do not feed later weeks' history.

6.7 Other uses

- **Aggregate price elasticity**, computed exactly through the base model by a fourth-order central difference of expected basket size in a common log-price change.
- **Customer-level questions** such as which households respond to a product's discount, from the household × product price slopes.
- **A benchmark law.** Synthetic data drawn from a fitted law, with known parameters, for testing other estimators.

7. Evidence and limits

The exact parts are verified, and on synthetic worlds with a known truth the model beats SHOPPER on likelihood and realism and makes correct discount decisions; on Dunnhumby it fits held-out baskets better than the base model but still overstates the discount response and adds a small gain to the rest of the basket.

7.1 Exactness

- **46 tests** check log Z, purchase probabilities, exact draws, Gibbs chains, importance sampling, pseudo-likelihood and the single-utility tilt against enumeration of every basket of a small world, for six model variants. They include a case where basket weights of different sizes differ by more than a double can hold.
- **An independent log-space calculation** on 64 real trips at chain-end z agrees with the fast code to within 1.5×10^{-10}, the per-trip cap's tolerance.

7.2 Synthetic stress world (known causal truth)

Measure	Version 4	SHOPPER (K = 8)
Held-out log-likelihood per basket	−24.66	−28.04 (about 3.0 behind once its ordering-estimate bias is allowed for)
Realism AUC, full basket / categories (0.5 is perfect)	0.62 / 0.73	0.92 / 0.89
Price scenarios, correlation with truth: own / same sub-category / rest of category	0.88 / 0.51 / 0.92	0.82 / ≈0 / ≈0

The own-price response has the right pattern but is about 1.7× too strong on average (+0.086 against +0.050).

Promotion decisions on a synthetic discount world (48 categories, seed 3, where discounts mostly move shoppers within a category; revenue relative to no promotion):

	10% off all 12	20% off all 12	30% off all 12	Correct decisions (of 12)
Truth	−3.4%	−7.1%	−10.8%	—
Version 4, latent state fixed	−3.4%	−6.8%	−10.2%	12
Version 4, latent draws reweighted	−3.0%	−6.0%	−9.0%	12

The base model alone (no competition term) made all 36 single-category cuts look profitable on three seeds of this world. Only one seed has been fitted with the newest settings.

7.3 Dunnhumby

Earlier model (co-purchase partition, test weeks opened once). Test log-likelihood −42.456 per basket against −46.097 for its base model: +3.64, household-clustered 95% interval [3.42, 3.87]. The gain grows with basket size: −0.09 for 1–3 products, +2.5 for 4–8, +7.3 for 9–20, +16.4 for 21–120.

Current model (commodity partition, before the normaliser fix), on validation:

Measure	Model	Data
Validation log-likelihood (1,024 trips)	−37.889	—
Discount depth slope	1.51	0.64 validation (95% interval 0.35–0.93); 1.07 train
Display effect, purchase-weighted	0.363	0.375
Mailer effect, purchase-weighted	0.383	0.347
Rest-of-basket effect of display elsewhere (items per trip per 100 displayed products)	+0.088 ± 0.023	−0.148 ± 0.025
Basket size: train early / train late / validation / test	5.59 / 9.79 / 6.60 / 7.39	5.50 / 10.77 / 8.33 / 8.75

The two likelihoods (−42.456 and −37.889) are on different trip sets and partitions and are not comparable. The current model's test weeks have not been opened.

7.4 Known weaknesses

- **The discount response is too strong.** The depth slope is 1.51 against 0.64–1.07 in the data, so discounts look more profitable than they are.
- **A halo remains.** Display elsewhere in the store raises the rest of the basket in the model (+0.088) and lowers it in the data (−0.148). Earlier models had +0.37.
- **Basket size is too small in later weeks** (6.60 against 8.33 on validation).
- **The size term and the item cap.** In the current fit, $\rho_o(m)$ rises to m = 9 and then falls to −26 at 39 categories and −273 at 120, and 130 of 187 categories have $\rho_c \le 0$. Gibbs chains started at observed baskets put 1 or 2 of 64 trips at the 120-product cap within 400 sweeps.
- **That fit was trained with a normaliser error.** At chain-end z, the old code underestimated log Z by 55–113 nats on 5 of 256 trips (2%), which training then used. It never happened at z near the observed basket or on synthetic data. Both fixes landed on 2026-10-26 and a refit with them started at 21:14; whether they explain the size-term shape is not yet known.
- **Training is slow.** One SML step on 256 trips takes about 5.5 s, mostly the gradients of the utilities and of the exact size table; a full fit took about 15 hours.

- **Within-category sums can drop terms** when the products inside one category span more than about 745 nats. A related error was found and fixed on 2026-10-26: the training path centred every category at the trip's top product, which lost up to 62 nats of log Z on about 5% of real trips at chain-end z; it now centres each category at its own top product, as the fast path did.
- **Not yet done:** the promotion environment on real data with a believable model, SHOPPER on Dunnhumby, and synthetic checks of the newest settings on more than one seed.

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This project (branch `v4-final` of the energy_basket_intelligence repository): `docs/THEORY.md` (derivations), `docs/PIPELINE.md` (stages and results; section 7.0 for the commodity model and the normaliser fix), `docs/STRESS_WORLD.md` (synthetic world), `v4/` (model, exact normaliser, sampler), `v4/competition_iv.py` (the λ_c instrument), `tests/test_law.py` (enumeration tests).